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RESEARCH ARTICLE

Privacy-Preserving ESG Risk Prediction via Federated Hybrid Neural-Boosting Ensembles

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ABSTRACT The rapid evolution of sustainability legislation, including the CSDDD and CSRD, has created a complex dual mandate for modern enterprises. Organizations are now required to provide granular transparency into their supply chains while simultaneously protecting proprietary operational data and trade secrets. This regulatory landscape exacerbates a critical financial problem: pervasive information gaps in ESG data hinder accurate risk assessment, especially for banks and investors who lack access to confidential supply chain details. Existing centralized reporting models are inadequate, as they either compromise data privacy or fail to deliver the cross-institutional insights needed for compliance. This paper addresses this challenge by introducing a privacy-preserving framework that combines a Hybrid Neural-Boosting Stack (HNB-Stack) with a decentralized Federated Voting-Ensemble System (Fed-VES). We employ a rigorous hybrid feature selection process to isolate high-value predictors from over 400 variables, which then inform a stacked ensemble architecture comprising LightGBM, XGBoost, and Multi-Layer Perceptrons. The results demonstrate that the HNB-Stack achieves superior predictive fidelity with a centralized R^2 of 0.9835, significantly outperforming traditional baselines. Most notably, the federated implementation maintains this high accuracy across five isolated institutional clients with minimal performance degradation, successfully proving that collaborative intelligence does not require centralizing sensitive data. By integrating SHAP and LIME for post-hoc interpretability, the framework also ensures that AI-driven risk assessments remain transparent and auditable. This research ultimately provides a scalable solution for navigating the regulatory landscape, enabling secure and reliable ESG reporting without compromising data sovereignty.

INDEX TERMS Federated learning, ESG risk prediction, stacking ensemble, privacy-preserving AI, explainable AI (XAI), sustainable finance, supply chain transparency.

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I. INTRODUCTION

Businesses currently operate in a highly globalized environment where sustainability laws are increasingly enforced

across most countries. These legislations demand that companies be more transparent in all their business activities. In the past, Corporate Social Responsibility (CSR) was mainly used to augment corporate reputation [1]. However, new standards such as the Corporate Sustainability Due Diligence Directive (CSDDD) and the Corporate Sustainability Reporting Directive (CSRD) mandate compulsory operational standards. Environmental and human rights compliance obligations are also being extended by national laws [2], [3]. Countries such as Germany, the United States, and France have instituted stern statutory compliance systems. Such systems compel companies to gather and validate sensitive organizational data, including labor policies, environmental impacts, and internal regulations. This presents a challenging, two-fold task for banking institutions. To meet regulatory expectations and investor requirements, banks must thoroughly assess Environmental, Social, and Governance (ESG) risks while simultaneously safeguarding client confidentiality and proprietary financial data. The traditional centralized pooling of ESG data amplifies breach and competition risks.

These requirements assist in governance and adherence, but present new confidentiality dilemmas. They generate privacy issues within organizations that are in an international regulatory setting. There is centralized data collection of vendors, which need to share highly sensitive information regarding operational data. The sellers can divulge trade secrets or competitive production powers on cloud-based systems. Firms are reluctant to provide granular ESG metrics that depict proprietary supply chains. These revelations may reveal operational effectiveness or strategic weaknesses to the competitors. This worry is even more in those industries where optimization of supply chain is of significance. This conflict becomes more obvious with CSDDD and General Data Protection Regulation (GDPR) limitations. CSDDD demands comprehensive supply chain audits in the operations of global business. The GDPR does restrict any cross-border transmission of identifiable or sensitive information. Fed-VES proposed by us offers a scalable and regulation-independent technical platform. It allows a joint ESG analysis without concentrating sensitive vendor data. This approach reduces ESG information asymmetry while preserving confidential operational data.

These mandates provide benefits to organizations in terms of governance and compliance; however, they create unique confidentiality and privacy issues across all of the organizations' international operations. With centralized processes for collecting data from vendors, there is an obligation for those vendors to provide highly confidential operational data, including trade secrets or competitive production abilities, on a cloud-based platform. Therefore, as vendors may need to provide accurate data to enable regulatory compliance with ESG, there may also be increased risk of having proprietary manufacturing processes exposed and having the company's proprietary/wage characteristics.

Many companies face additional burdens to live up to the requirements of the EU's General Data Protection Regulation (GDPR). The GDPR limits how companies can process information and transfer it across international borders, especially when it is commercial in nature. Therefore, many companies face both a regulatory requirement for transparency regarding data processed by outside parties as well as a need to maintain their customers' confidentiality and promote the sovereignty of their respective countries regarding data. This conflicting obligation makes it difficult for accounting professionals to ensure that the information they use is of high quality and useful when making business decisions. To have reliable and credible sources of data in relation to ESG, it is essential for users in the Accounting field to have access to ESG data in a timely, complete and impartial manner. Unfortunately, many vendors of ESG data are reluctant to provide end-users with access to their proprietary data sets for this purpose, thus limiting the end-user's ability to obtain a comprehensive understanding of the risks that exist to both the environment and society and to assess how well their organization manages those risks. In addition, as with financial reporting, there is not currently a standard framework for developing and reporting ESG data; therefore, any ESG information contained in a user's current accounting system will not be accurate unless it has been subject to extensive privacy protections. Due to the absence of a standardized ESG data framework, companies will be required to create and deploy effective governance frameworks that will conserve the secrecy of their sensitive ESG data and at the same time make them able to meet regulatory obligations. The presence of these mechanisms is required in order to develop credible, accountable and decision supportable ESG reports in a multifaceted international business.

The existing models of ESG forecasting rely largely on centralized aggregation of data that reveals sensitive information and creates significant differences in the reporting framework due to different reporting demands. Moreover, the majority of the existing models are concerned with separated metrics; thus, the findings of such models cannot be readily generalized and cannot be understood. The research presents a new method of privacy-preserving forecasting that is developed on the Hybrid Neural-Boosting Stack combined with a Federated Voting-Ensemble System. The HNB-Stack has several underlying learners such as LightGBM, XGBoost and MLP, combined via a Bayesian Ridge meta-learner and augmented with SHAP and LIME which give it explainability. With the multi-stage preprocessing and hybrid feature selection, HNB-Stack was 1.9281 with an RMSE and 0.9835 with an R . The RMSE of Fed-VES was 2.0696 and R^2 was 0.9809 with five clients, which guarantees the data sovereignty and offers auditory information on the performance metrics.

- The HNB-Stack captures nonlinear hierarchies in ESG-financial data, reducing RMSE by 13.8% and increasing R^2 by 4.2% over CatBoost.

- Hybrid feature selection isolates key predictors from 400+ variables, improving stability and reducing multicollinearity.
- Fed-VES enables decentralized learning without sharing raw data, achieving near-centralized performance with R^2 loss below 0.3% and supporting regulatory compliance.
- XAI via SHAP and LIME provides transparent explanations of ESG drivers for investors and regulators.
- The framework explains 98.35% of variance centrally and 98.10% federated, improving over single-learner models in handling outliers and non-stationary data.

The database analysis comprises eight sections. The next section discusses previous studies relating to ESG fulfilment, privacy risk factors, and how federated education fits into these studies; then the following section explains how the sample data was collected and cleaned. The sample variables were recorded, and the HNB-Stack and Fed-VES models were stacked based on the sample variables. The last section presents the results of all test evaluations for the HNB-Stack and Fed-VES; then the next section contains a discussion of limitations of the current study; finally, the sixth section includes a summary of how the results of both studies have implications for real and academic settings. The seventh section presents policy and managerial implications, and the eighth section provides the conclusion.

II. BACKGROUND AND RELATED RESEARCH

A. ESG DISCLOSURE REQUIREMENTS AND PRIVACY CHALLENGES

The Corporate Sustainability Due Diligence Directive European Union is an important milestone in the evolution of global sustainability governance because it will establish new ways for organizations to address the dual challenges of compliance with their increasing requirements for transparency, as well as the equally stringent requirements for privacy protection. The effective date of the CSDDD is July 25, 2024. As of this date, CSDDD will impose legally enforceable duties on all corporations that are as of this date, CSDDD will impose legally enforceable duties on all corporations required to trace human rights and environmental damage throughout extended global supply chain European Union including any obligations under existing European Union law and regulations [4], [5]. Additionally, the CSDDD requires climate transition plans that fulfill the Paris Agreement's conditions for limiting warming to 1.5°C, and will also create obligations to monitor regularly the following indicators: deforestation, hazardous-waste spilling and frequency of lost-time injuries [6].

As new requirements place additional obligations on social responsibility reporting, they highlight how these increased expectations contain social responsibility reporting within a broader framework that is more politically oriented and accountable than traditional compliance. In addition, Ströher et al. [1] emphasise how the exchange of carbon

data in an automated way between organisations requires a massive redesigning of existing systems to enable privacy protection. Finally, Baryannis et al. [7] also discuss how the ongoing issues of obtaining, verifying, and aggregating data have increased due to the introduction of full-scale regulations, such as the CSDDD.

Mounting implementation pressures that can affect environmental sustainability, the rising costs associated with meeting compliance obligations, and heterogeneous methods used in measuring environmental sustainability can lead to fragmentation within the industry, as noted in industry assessments [3]. These problems are made worse because the many ESG systems overlap and create a fragmented landscape. For example, there are lots of Global Reporting Initiative measures for environmental, social, and governance issues. Also, the Sustainability Accounting Standards Board has different measures for ESG issues in certain fields. And there are the International Sustainability Standards Board standards [8]. Studies show that it's hard to compare non-financial measures and to combine different ESG measures into one reporting system [9], [10]. The escalation of enforcement activity in the area of climate disclosures and ESG-related violations by federal agencies (including the SEC and the U.S. Department of Justice) in 2024 has increased the importance of developing privacy-preserving compliance solutions that enable organizations to meet the growing regulatory obligation while protecting the commercially sensitive information associated with their ESG practices.

B. PRIVACY-PRESERVING TECHNOLOGIES FOR ESG COLLABORATION

Privacy-enhancing technologies play a key role in helping organizations to meet the regulatory duality of transparency versus confidentiality being created by recent ESG regulations. Under the CSDDD regulations, for example, organizations are encouraged to implement digital compliance tools to meet their disclosure requirements, while protecting their trade secrets and owner confidentiality [4]. This approach to solving this current regulatory issue is consistent with the technical guidance for the EDPS issued in 2025, which reiterated that a federated learning framework minimizes privacy risk by distributing the source (raw) data throughout the system, thereby making it appropriate for use in situations where a centralised data repository would violate regulatory, ethical, or contractual limitations on the availability of data [11].

As a result, Federated Learning (FL) has developed into a widely popularized decentralized method of Machine Learning, where a number of different organizations can work together to train common models without having to share or otherwise expose underlying datasets with each other [12]. FL has resolved a long-standing issue that has existed in many organizations, which has been the ability to build collective analytical skills while keeping individual

competitive boundaries intact. Liu et al. [13] suggest that businesses should exchange model updates or other less valuable information associated with traditional operations in order to obtain higher-value information, for example through federated data exchange. Their adaptive federated learning architecture shows how the creation of an artificial intelligent agent that operates in a decentralized manner promotes data sovereignty and meets the current expectations of due diligence outlined in many newly created regulatory frameworks.

Practical use cases provide confirmation of the practicality of these technologies. The IBM Federated platform enables retailers to assess performance across multiple locations, while at the same time enabling manufacturers to synchronize geographically separated operational activities while also keeping proprietary transaction data secret. Additionally, Zhu et al. [14] provided a compelling contribution to the practicality of these new industrial innovations by combining trusted execution environments with blockchain verification in order to enhance security and privacy in the current federated infrastructure, demonstrating how multiple privacy-preserving technologies can be effectively integrated into complex business networks.

C. FEDERATED LEARNING APPLICATIONS IN ESG MANAGEMENT

In the field of intricate production and distribution systems, Federated Learning has been shown to be of significant value as a method of developing a collaborative brain to assist the requirements of organisations without encroaching on their privacy. Zheng et al. [15] show that even groups with not much data can still build a good risk prediction model if they use decentralized models. Their work shows how much you can improve by working together, because it gives groups access to shared information while keeping their private data safe. Also, their study of order delays before and during Covid-19 shows that things like uneven data, how bad the disruption is, and which algorithm you pick all really change how well a model works. As such, the need for sufficient federated configurations in a real-world environment cannot be overstated.

In their 2025 research, Tang et al. build on this research with an Isomorphic Federated Graph Neural Network, to fill the gaps from long-standing visibility issues created by strict confidentiality requirements that stop organizations from sharing complete raw data sets. Their model uses link prediction and structured data transfer to help organizations conduct thorough ESG Risk Assessment and report findings to their stakeholders without compromising on data privacy and security. Further, a recent systematic review backs up this statement and found that many companies operate from incomplete data sets and encounter large-scale privacy challenges when developing collaborative models of risk across multiple stakeholders [16]. Federated Learning enables groups to analyse and utilise data without the need to share sensitive data while maintaining the highest level

of security and compliance with existing regulations. ESG-related risk contributes to all levels of risk within the supply chain continuum of a company. When a company fails to adhere to the ESG principles, whether by its activities in the environmental realm or how it treats employees within the supply chain continuum, the company is likely to receive negative media coverage, which can adversely affect its public perception. Unlawful problems may also be brought to companies that may be linked to their suppliers due to poor governance [17]. Quality data is highly essential in any risk management initiative; Amjad et al. [18] also support this notion by saying that quality data is essential to AI as a risk management tool. The use of Federated Learning is a perfect approach to enhancing the gathering and storage of the quality data required in the competent management of risks because of its confidentiality.

The examples of cross silo applications can give an idea of the way in which this technique could be implemented in the real world situations. Among them is the article by Schreyer and others, published in the journal of Accountancy, in 2022 [19]. The auditors further relied on various financial source data to determine the level of risk of fraud and ensure confidentiality in the determination of the occurrence of fraud with 98% success. Also, they demonstrated how their practice could be used as a model of Environmental, Social, and Governance (ESG) management. Some organizations, however, nevertheless encounter certain limitations. As highlighted by Zheng et al. [15], organizations that possess large amounts of internal datasets may derive minimal incremental value due to a lack of incentive alignment examples within their specific contexts. In addition to the above, Zhang et al. [20] summarize other practical issues that organizations may experience, including data heterogeneity and interoperability issues. Both Kong et al. [21] and Hamidu et al. [22] provide recent contributions to this topic by introducing order-level frameworks for predicting risk and deep-learning implementations to support operational resiliency, respectively.

D. RECENT METHODOLOGICAL FRONTIERS IN AI-DRIVEN SUSTAINABLE FINANCE

Recent advancements in AI-driven sustainable finance have introduced highly sophisticated paradigms, notably differentially private federated optimization [23], [24], graph-based ESG modeling [25], and causal machine learning [26], [27]. Differentially private federated learning (DP-FL) provides rigorous mathematical guarantees against gradient leakage through bounded noise injection; however, it often does so at the cost of severe convergence instability and degraded model utility when applied to highly heterogeneous financial datasets [23], [24]. Simultaneously, graph neural networks (GNNs) have emerged as powerful tools to model multi-hop ESG risk contagion across complex global supply chains, effectively capturing inter-firm relational dependencies that traditional firm-level models overlook [25]. Furthermore, causal machine learning techniques, such as Double Machine Learning (DML) and Causal Forests, are increasingly being

deployed to estimate the true, heterogeneous treatment effects of green interventions, moving the field beyond mere correlational forecasting [26], [27].

While theoretically profound, these advanced methods frequently encounter practical adoption barriers in heavily regulated banking environments due to massive communication overhead, loss of granular interpretability, or strict observational assumptions. In contrast, our proposed HNB-Stack and Fed-VES architecture delivers a critical applied innovation: it deliberately eschews heavy topological complexity and noise-injected gradients in favor of a mathematically transparent, computationally lightweight ensemble optimized specifically for tabular corporate disclosures. By bridging the gap between advanced collaborative intelligence and pragmatic regulatory auditability, this framework provides a highly scalable, immediate solution for financial institutions requiring near-centralized predictive fidelity without sacrificing operational interpretability.

III. METHODOLOGY

In this section, the authors discuss the design of their feature-optimized stacking ensemble for estimating ESG performance, that of conducting rigorous preprocessing with hybrid feature selection, along with the use of trusted AI throughout the one-stage integrated process. They used the data of state-owned enterprise-related features of the environment and finance, which underwent multi-level cleaning of data. This is carried out by the elimination of features showing less variability, imputation of the median for numerical features, imputation of the classes for categorical features, and normalizing by using the RobustScaler for elimination of the effects of the outliers. The authors design their Hybrid Neural-Boosting Stack using the F-Regression, the Mutual Information, along with the Random Forest, where it retained the top twenty features. They also used LightGBM, XGBoost, along with the Multi-Layer Perceptron Neural Network. For privacy considerations, they also used Federated Learning for five institutional customers. They used various measures for the model evaluation, along with the interpretation for the AI evaluation. They used SHAP and the LIME for interpretation. The overall methodology of our proposed framework is depicted in Fig. 1

A. DATA AND SAMPLE

This study mixes environmental, social, and governance info with money data from big state-owned companies. The info includes years of data about company environmental actions, social responsibility, company leadership, and money results. The dataset was divided into a training set (80%) and testing set (20%) to maintain the integrity of the data. Columns that were not important or didn't have enough variance were removed as were rows where more than 40% of the values were missing. The missing values were replaced with their median value and the mode for categorical data. Categories were transformed into numeric codes using the LabelEncoder and the ordering of the categories was preserved where

appropriate. Finally, the dataset was normalized using robust scaler which is ideal for scaling data containing outliers. This cleaned-up data is then used to pick the best features before we train and test our Hybrid Neural-Boosting Stack and Federated Voting-Ensemble System.

B. DATA PREPROCESSING

To get the data ready, we follow a simple process that keeps the time order correct and stops data from the test set affecting the training set. We split the data 80/20. We clean up the data by removing columns that are IDs, don't change much, or are mostly empty. For number gaps, we fill them with the middle value because that works better when the data is uneven. For categorical gaps we fill them with the value that occurs most frequently, using a LabelEncoder for consistency. We standardise features using RobustScaler to handle outliers. After standardising, we select our Top 20 features by combining rankings from F-regression, Mutual Information, and Random Forest importance. These best attributes are the input of the Hybrid Neural-Boosting Stack, based on which we make predictions of ESG scores and justify them with the use of SHAP and LIME.

C. VARIABLE MEASUREMENT

Triad approach isolates the most important predictors in the case of choosing a feature set out of the enterprise data. F-regression is a simplistic linear style of mapping features to the target, Mutual Information illustrates nonlinear correlations, and random forest measures the value of each feature in a tree to decrease the noise in a prediction. When these outcomes are integrated, the combined methodology gets down to the bottom of what motivates the environmental sustainability of companies, their economic profitability, governance, performance, and social responsibility. The measurement of the sustainability performance is a rounded perspective by governance scores, metrics of board effectiveness, metrics of social responsibility, and the quantitative measures of normalized revenues, resource usage, emissions, and working capital. Repeated analysis shows this feature set distinguishes high and low ESG performers and fits the Hybrid Neural Boosting Stack and Federated Voting Ensemble System.

The operationalization of ESG risk in this study is theoretically grounded in Stakeholder Theory and Institutional Theory. Rather than treating ESG performance as a sheer numerical regression target, we define our target variable `corpgov_tresgscore` as a comprehensive, retrospective composite score that quantifies an organization's alignment with coercive regulatory pressures under Institutional Theory and its capacity to manage competing multi-stakeholder expectations under Stakeholder Theory.

This target variable is constructed as an aggregated historical sustainability performance metric, harmonizing granular indicators across environmental emissions, workforce dynamics, and corporate governance architectures in accordance with established reporting frameworks like the Global

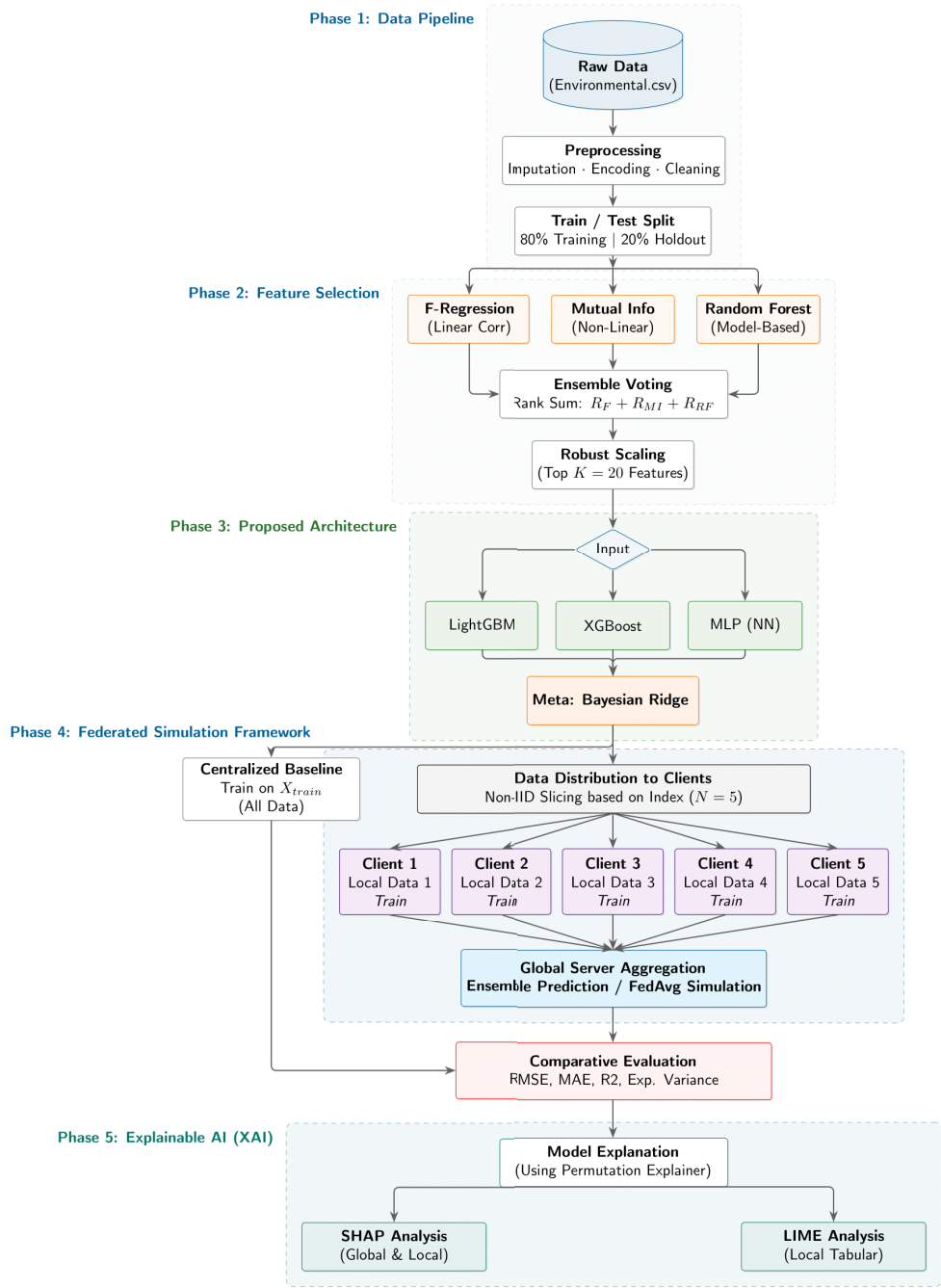


FIGURE 1. The proposed methodological pipeline.

Reporting Initiative and SASB. By predicting this composite metric, the proposed HNB-Stack transitions ESG risk from an abstract conceptual compliance burden into a quantifiable, audit-ready operational benchmark, providing stakeholders with a mathematically grounded gauge of historical institutional resilience and latent risk exposure.

D. PREDICTIVE FEATURE ENGINEERING AND SELECTION

Complementary methods are used for a three-way hybrid feature selection to sift through over 400 Environmental, Social, and Governance (ESG) variables. F-regression picks

up linear ties, Mutual Information unmaskes non-linear dependencies, and Random Forest importance gauges how features contribute within the hierarchy. Each method converts its rankings into percentile scores and combines the normalized rankings into a consensus relevance score to reduce single-method bias. An automated iterative loop trains light stacking models on growing five-feature subsets using nested cross-validation to determine the best feature count. Features selected are standardized with RobustScaler through interquartile-range normalization. This refined twenty-feature subset enhances forecasting accuracy

of the Hybrid Neural-Boosting Stack, provides stability, and amplifies interpretability while scaling effectively across ESG datasets and institutional settings.

E. NOVEL HYBRID FEATURE SELECTION STRATEGY

A combination of three voting-based selectors was used to sift the most informative twenty features out of the preprocessed pool, as shown in the figure on the correlation matrix after feature selection. The strategy is an integration of three complementary strategies: F-regression in the linear correlation between features and the target, Mutual Information in the non-linear correlation and complex interactions between the features, and Random Forest importance in the extent to which each feature can diminish the uncertainty in the trees. The total scores of each item are calculated as a sum of the three scores of all the selectors. The highest-scoring items are retained, and this prevents biases brought about by a single approach. Furthermore, while the individual algorithms comprising the hybrid feature selector are established, their composite integration serves a critical theoretical purpose, Stability Selection in non-stationary data. ESG disclosures are notoriously heterogeneous and subject to temporal drift. Relying on a single feature selection paradigm often leads to highly unstable subsets across different data folds. By enforcing a strict consensus across linear F-regression, complex non-linear Mutual Information, and hierarchical Random Forest dependencies, our voting mechanism effectively acts as a multi-dimensional stability filter. This ensures that the selected Top 20 predictors are not artifacts of local data anomalies, but possess consistent, theoretically grounded explanatory power across varied institutional reporting structures. The combined approach mitigates the risk of erroneously discarding informative features while preserving the diversity required for robust ensemble prediction, thereby supporting a more reliable ESG evaluation. This effect is clearly evidenced by the substantially reduced multicollinearity observed in the correlation matrix following feature selection, as shown in Fig. 2.

F. FEDERATED LEARNING SIMULATION AND GLOBAL MODEL AGGREGATION

A more realistic simulation framework for federated learning is a decentralized system that is privacy-conscious and governed by rules of governance and confidentiality. The Federated Learning Simulation Framework offers an accurate portrayal of a decentralized system and considers the privacy rights of users and the governance of the system. It also offers a means to maintain the confidentiality of all data stored within the system. The Federated Voting-Ensemble System was designed to illustrate how to develop knowledge that improves functionality when repositories cannot be shared, showing that privacy can remain in place. The central HNB-Stack model simulation framework assists in identifying the costs associated with the preservation of privacy in relation to the Fed-VES simulation framework.

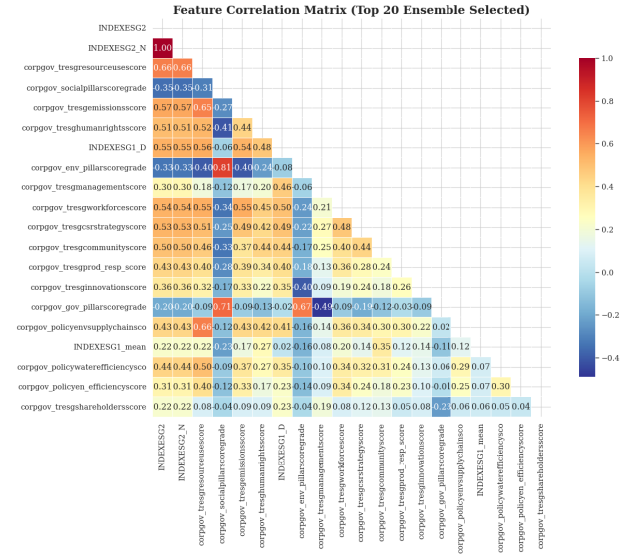


FIGURE 2. Correlation matrix after feature selection.

G. FED-VES: LIGHTWEIGHT CLIENT ARCHITECTURE

Fed-VES, short for Federated Voting-Ensemble System, is a simple framework that institutions can use on their own computers for decentralized work. It's built to keep computing and data transfer needs low. Each institution has a small Hybrid Neural-Boosting Stack with three efficient base learners: LightGBM, XGBoost, and a lean Multi-Layer Perceptron with one hidden layer. A BayesianRidge meta-learner combines the outputs using probabilistic weight estimation. The system limits model size so it can run on limited infrastructure but remain correct through ensemble diversity. Local training uses five-fold cross validation. Institutions then submit only test set predictions to a central server, keeping raw data private and enabling collaboration. Fed-VES supports growth through partnership across an enterprise ESG prediction system.

H. ESG PREDICTION PROTOCOL

The method by which the ESG Prediction Model generates uniform scores for each customer is based upon a simple process. Data must be prepared using the exact steps described in this document. When preparing data for scoring, we remove anything that could identify the source of the data and anything that will not have a significant impact on our calculation. Once missing numbers have been replaced by the average of the other values, we replace any missing categories with the most frequent category, keeping track of any newly added items. After these procedures, we convert the data to standard form using inter-quartile normalization prior to inputting it into the model

$$X_{\text{scaled}} = \frac{X - Q_1}{Q_3 - Q_1}. \quad (1)$$

The distribution of the data remains stable as training shifts to deployment. For structured prediction with the Hybrid Neural-Boosting Stack, twenty features were used.

Each base-learner's output is computed independently using LightGBM, XGBoost, and a calibrated multi-layer perceptron, with a Bayesian Ridge meta-learner combining these heterogeneous signals into one final prediction,

$$\hat{y} = w_1\hat{y}_1 + w_2\hat{y}_2 + w_3\hat{y}_3, \quad (2)$$

yielding robust institutional ESG score estimates. The Federated Voting-Ensemble System aggregates local predictions from five clients to compute a global consensus using

$$\hat{y}_{\text{global}} = \frac{1}{5} \sum_{i=1}^5 \hat{y}_i, \quad (3)$$

which enhances cross-institution reliability without violating data-isolation constraints. Post-prediction interpretability is achieved through SHAP and LIME, where feature-wise contribution analyses strengthen stakeholder comprehension and reinforce regulatory transparency.

I. FEDERATED LEARNING FOR MODEL TRAINING

Federated learning allows five institutions to collaboratively develop a shared model while keeping data strictly confidential during the full training process. Each client k trains the Hybrid Neural-Boosting Stack locally on its own dataset D_k , never sending raw samples outside of its secure environment. Within their local setup, the client optimizes LightGBM, XGBoost, and Multilayer Perceptron learners by minimizing the local objective,

$$L_k = \frac{1}{n_k} \sum_{i=1}^{n_k} \text{MSE}(\hat{y}_i, y_i), \quad (4)$$

where n_k denotes the number of local training samples. A five-fold internal cross-validation scheme mitigates overfitting risks, producing a generalization estimate defined as

$$\text{CV}_{\text{error}} = \frac{1}{5} \sum_{j=1}^5 L_{\text{test},j}. \quad (5)$$

Each client sends only its model parameters θ_k after local training to the central aggregation server so that sensitive institutional data remains private. Then, the global parameter fusion is performed by the server using

$$\theta_{\text{global}} = \frac{1}{K} \sum_{k=1}^K \theta_k. \quad (6)$$

With $K = 5$ in total, the participating institutions share back a consolidated global model. This configuration is capable of sustaining iterative refinement in a series of numerous rounds of communications. Finally, the predictions of individual clients are aggregated to form one, unified estimates of ESG using ensemble averaged by federated

$$\hat{y}_{\text{fed}} = \frac{1}{K} \sum_{k=1}^K \hat{y}_{k,\text{test}}. \quad (7)$$

Such a federated training protocol provides strict privacy guarantees at predictive performance that is similar to centralized model training.

J. TRAINING PROCESS

The training process has a very clear order: preprocessing of data, construction of base models, and fine-tuning of the meta-learner. Under and across all institutional clients the ESG and financial data are subjected to the same preprocessing: low-variance columns and identifiers are eliminated, gaps between the missing numeric variables are filled with the median and gaps between the categories with the most common category assigned. RobustScaler scales features. The ensemble is fed by the twenty hybrid-ranked features. Five-fold stratified cross-validation retains institutional and temporal structure. Base learners are trained on four folds and tested on one. BayesianRidge is a meta-learner that takes a mix of predictions, with emphasis on the more powerful models, using out-of-fold predictions. The training is conducted independently in five institutions in federated learning that keeps data confidential.

K. CROSS-VALIDATION

Five-fold cross-validation is a clear and fair means of measuring the performance of the stacking ensemble that is sector-agnostic. Training data is broken down into five folds. Base learners- LightGBM, XGBoost and a Multi-layer perceptron, are trained on four folds and the remaining fifth is used to validate the learners. The model learning is also obtained with out-of-data estimates. BayesianRidge combines base learner prediction in a Meta Learner, with probabilistic weight estimation and an in-built form of regularization to curb overfitting. This repetition lowers overfitting and smooths industry volatility and takes into account ESG disclosure discrepancies. Measures such as RMSE, MAE, R^2 , MedAE, MAPE, and MdAPE confirm a good generalization and predictive stability.

L. HYPERPARAMETER OPTIMIZATION

The ensemble methodology suggested has a major degree of incorporating hyperparameter tuning to minimize empirical risk, and makes performance consistent between organizations and reporting periods. The framework is based on transparent, tree-based ensemble algorithms with probabilistic regularization, purposely avoiding deep learning to reduce complexity and enhance interpretability. The tuning process targets the Light Gradient Boosting Machine (LightGBM) (learning rate, maximum depth, minimum number of samples per leaf) and eXtreme Gradient Boosting (XGBoost) (λ regularization, γ regularization). BayesianRidge tuning is used to control coefficient variance by tuning α . Five-fold stratified cross-validation is used to evaluate the model to mitigate the risk of overfitting on sparse and multimodal ESG data, ultimately minimizing the mean-squared prediction error.

$$\theta^* = \arg \min_{\theta} \sum_{k=1}^K \text{MSE}_k(\theta), \quad (8)$$

where K denotes the number of folds and MSE_k represents the validation error for fold k . The aggregated cross-validated loss is computed using

$$CV_{\text{loss}} = \frac{1}{K} \sum_{k=1}^K L_{\text{test},k}, \quad (9)$$

which provides an unbiased performance estimate that is resilient to the in-sample distortions. The well-defined objective function yields a robust optimization module that can absorb variability in ESG reporting and the inconsistencies of the institutional disclosures among the collaborating clients.

1) MODEL-SPECIFIC OPTIMIZATION ARCHITECTURES

Optimizing different base learners involves understanding their structure and requirements. Gradient-boosting models like XGBoost and LightGBM use a hybrid search strategy combining grid-search and random variations to find optimal hyperparameters. The model probes learning rates from 0.01 to 0.1, tree depths from 3 to 8, and subsample rates of 0.6–1.0. Regularization limits variance while allowing some non-linearity. The compact multilayer perceptron has hidden layers with 64 and 32 neurons, using ReLU activation and L2 penalties between $1e-4$ and $1e-2$. Training stops if validation error no longer improves. BayesianRidge meta-learning uses log-based alpha grids and k-fold cross-validation, stabilizing weight estimation and developing a stacked regression model from out-of-fold predictions,

$$\hat{y} = \sum_{j=1}^M w_j f_j(x; \theta_j), \quad (10)$$

which is optimized through a convex penalized objective,

$$\min_w \left[\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda (w^\top w) \right]. \quad (11)$$

From a theoretical perspective, the aggregation mechanism of the HNB-Stack represents an advanced implementation of optimum linear soft fusion. Recent analytical developments in ensemble fusion emphasize that the performance of a linear combination of models heavily depends on the cross-correlation of their outputs. Let $\mathbf{z} \in \mathbb{R}^K$ denote the vector of continuous predictions from the K base learners, and y denote the true ESG score. The goal of optimum linear soft fusion is to find the weight vector $\mathbf{w}$ that minimizes the expected mean squared error, $\mathbb{E}[(y - \mathbf{w}^\top \mathbf{z})^2]$. Analytically, the unregularized optimal solution is governed by the Wiener-Hopf equation:

$$\mathbf{w}^* = \Sigma^{-1} \mathbf{c} \quad (12)$$

where $\Sigma = \mathbb{E}[\mathbf{z}\mathbf{z}^\top]$ is the $K \times K$ covariance matrix of the base learners' predictions, and $\mathbf{c} = \mathbb{E}[y\mathbf{z}]$ is the cross-correlation vector between the predictions and the target.

This theoretical formulation is critically important from the point of view of our specific application. Corporate ESG data is intrinsically noisy, incomplete, and subject

to institutional reporting inconsistencies. Because our base learners (LightGBM, XGBoost, and MLP) are trained on the same underlying multimodal data, their residual errors are highly correlated, rendering the covariance matrix Σ ill-conditioned. Under these application-specific conditions, the unregularized matrix inversion Σ^{-1} amplifies minor data fluctuations, leading to volatile fusion weights and severe overfitting.

To overcome this analytical bottleneck, the HNB-Stack utilizes a Bayesian Ridge meta-learner, which fundamentally transforms the aggregation into a probabilistically regularized optimum soft fusion. By imposing a zero-mean Gaussian prior on the weights, $p(\mathbf{w} | \alpha) = \mathcal{N}(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I})$, and assuming Gaussian noise in the predictions with precision β , the optimal fusion weights are mathematically redefined by the maximum a posteriori (MAP) estimate:

$$\mathbf{w}^* = (\Sigma + \alpha/\beta \mathbf{I})^{-1} \mathbf{c} \quad (13)$$

This analytical development introduces a diagonal loading factor ($\lambda = \alpha/\beta$) directly into the covariance matrix. From an application standpoint, this mathematical regularization ensures that the matrix inversion remains stable despite the extreme multicollinearity of the ESG base predictions. The fusion process inherently penalizes redundantly correlated models while optimally allocating weights to independent predictive signals. Consequently, this theoretically grounded soft fusion mechanism directly accounts for the superior predictive fidelity and robustness of the HNB-Stack when navigating the volatile landscape of global ESG risk disclosures.

The configuration used to create the HNB-Stack and all book models was very useful. This study utilized five-fold stratified cross-validation, giving each institutional sector and period its own data split. Candidate predictor variables were selected by combining three feature engineering methods: F-Regression, Mutual Information, and Random Forest importance ranking. These outputs provided a shortlist of the 20 best candidate predictors with substantial and consistent explanatory power in the rapidly changing, non-stationary ESG environment. This methodology enhances generalization and reduces overfitting across diverse market conditions, as summarized in Table 1.

M. MACHINE LEARNING ARCHITECTURES AND HYBRID NEURAL-BOOSTING STACK

This element targets tabular machine learning methods that are interpretable and relevant for combining ESG and Financial data, where temporal patterns may be less distinct and having transparent, easy-to-understand decision-making logic is more important. The methods include gradient-boosted trees (GBT), probabilistic linear regression (PLR), and compact neural networks (CNN), each of which has its unique inductive strengths. Each individual learner has been validated using independent five-fold cross-validation, resulting in out-of-fold predictions that can be input to the meta-learner. The use of GBT helps refine an overall

TABLE 1. Model configuration summary.

Model	Hyperparameters	Notes
XGBoost	learning_rate=0.03, max_depth=5, n_estimators=500, subsample=0.8, colsample_bytree=0.8, reg_lambda=1.0	Reliable gradient boosting with strong nonlinear capacity
LightGBM	learning_rate=0.02, num_leaves=31, n_estimators=600, feature_fraction=0.8, bagging_fraction=0.8	Efficient boosting method offering rapid training and high accuracy
MLP Regressor	hidden_layer_sizes=64-32, activation=relu, alpha=0.0005, max_iter=500	Neural architecture capturing smooth nonlinear interactions
Bayesian Ridge	alpha=auto, normalize=True, n_iter=300, tol=1e-3	Probabilistic meta-learner enforcing adaptive Bayesian regularization
Cross-Validation	KFold(n_splits=5, shuffle=True, random_state=42)	Unified evaluation for base and meta-level models
Feature Selection	Hybrid ranking: F-regression, Mutual Information, Random Forest → top 20	Curated subset delivering stable and interpretable predictors

predictive model through an iterative process that involves combining weak predictors with additive updates at each stage,

$$F_m(x) = F_{m-1}(x) + \eta h_m(x). \quad (14)$$

In an approach to training via gradient descent, we can approximate negative loss gradients with “stumps” (i.e., shallow trees) to build up a set of weak learners with limited ability to generalise. To account for noise in an ESG indicator, we can control the learning rate (η) over time. As we noted, both XGBoost and LightGBM utilise L1 and L2 regularisation along with histogram leaf growth to find non-linear partitions where the sector has the greatest impact on the model predictions. Multilayer perceptron structures are built using a restrictive architecture to find a balance between providing a stable solution and maintaining limited computational complexity. The proposed transformation can be formalised in the following manner:

$$h^{(1)} = \sigma(W_1 x + b_1), \quad \hat{y} = W_2^T h^{(1)} + b_2, \quad (15)$$

where ReLU activation (σ) promotes efficient gradient propagation under L2 regularization. The architectural novelty of the HNB-Stack lies in its specific fusion of diverse inductive biases governed by a probabilistically regularized meta-learner. In the context of federated ESG prediction, standard meta-learning approaches often overfit to the predictions of the strongest local base learner, leading to severe generalization gaps during global aggregation. To counteract this, the HNB-Stack utilizes Bayesian Ridge regression as the meta-estimator. Unlike standard linear combiners, Bayesian Ridge introduces a formal probabilistic optimization principle by treating the ensemble weights as Gaussian distributions and

computing a posterior distribution over them. This imposes an embedded regularization and adaptive shrinkage mechanism, analogous to dynamic L2 regularization, which automatically penalizes overly confident or highly variant base-learner predictions. Consequently, this embedded sparsity mechanism guarantees theoretical consistency in the ensemble’s predictions, ensuring robust, stable convergence even when individual institutional clients provide noisy or heavily skewed ESG parameter updates. The neural-boosting hybrid stack uses a two-tiered, organized integration of predictive signals. Predictions from base learners create a meta-feature matrix,

$$\mathbf{Z} = \begin{bmatrix} \hat{y}_{1,1}^{(CV)} & \cdots & \hat{y}_{1,K}^{(CV)} \\ \vdots & \ddots & \vdots \\ \hat{y}_{n,1}^{(CV)} & \cdots & \hat{y}_{n,K}^{(CV)} \end{bmatrix}, \quad (16)$$

which synthesizes diverse model responses into consistent cross-validated representations. The stacked ensemble computes final predictions through

$$\hat{y}_i = \sum_{k=1}^K w_k \hat{y}_{i,k}^{(CV)} = \mathbf{z}_i^T \mathbf{w}. \quad (17)$$

Meta-optimization minimizes a regularized quadratic objective,

$$\min_{\mathbf{w}} \left[\frac{1}{n} \sum_{i=1}^n \left(y_i - \mathbf{z}_i^T \mathbf{w} \right)^2 + \lambda \left(\mathbf{w}^T \mathbf{w} \right) \right]. \quad (18)$$

By maintaining stable weights and reducing overfitting in noiseless high-dimensional spaces, this construction enables learners’ contributions to be equally divided while preventing a single method from being overly dominant in training. We utilise repeated K-fold stratified cross-validation with a random_state of 42 when tuning hyperparameters in five folds, allowing for tuning hyperparameters to be completed separately in each fold. The convex-hull performance of the resulting architecture provides robust predictive performance that is resilient against inconsistencies in reporting. In sum, the ensemble outperforms all individual base learners by providing a single stacking framework that provides stable, flexible, and probabilistically regularised performance.

IV. RESULTS AND DISCUSSION

A. MODEL PERFORMANCE AND FEATURE OPTIMIZATION OVERVIEW

Regression performance will be used to compare HNB-Stack against existing baseline methods. To do this, we will compare the performance of base learners that include LightGBM, XGBoost, and a MLP to other commonly used algorithms, such as SVR, RF, and k-NN. All models will be trained using subsets of features from over 400 features selected using a hybrid selection strategy, using five-fold stratified cross-validation patterns across three institutional sectors and three time periods. To reduce the ability to overfit

the model where ESG data is limited, the hyper-parameters will be tuned adaptively with an early stopping method (Fig. 3) to provide tighter fits to the true ESG scores as compared to the other models. The results demonstrate that the HNB-Stack effectively captures hierarchical governance structures and nonlinear sustainability relationships through the synergistic use of gradient boosting trees and a neural network, as evidenced by the close alignment between ground truth and predicted ESG scores shown in Fig. 3.

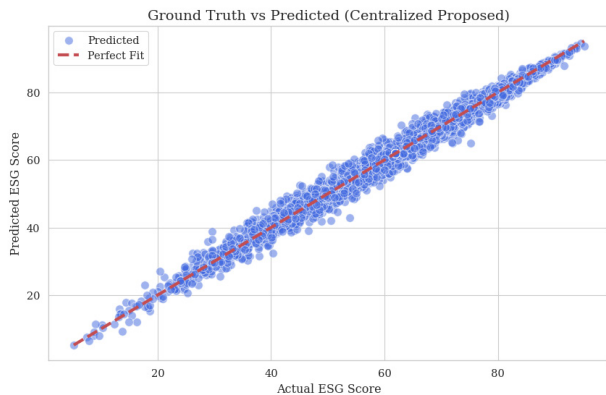


FIGURE 3. Ground truth vs Predicted for the ESG.

B. HYBRID NEURAL-BOOSTING STACK PERFORMANCE SUPERIORITY

Based on all available metrics for evaluation, the HNB-Stack displayed superior predictive performance compared to any other baseline model. The amount of RMSE for the HNB-Stack was 1.9281, which is considerably greater than that of either the individual base learners or any standard algorithms or methods. With an MAE of 1.3500, it is clear that the model is capable of accurately capturing every ESG score throughout its lifecycle. The R^2 value (0.9835) indicates that 98.35% of the variability in the dependent variable can be explained by this model, thereby providing near optimal levels of predictivity and explanatory power. The Explained Variance also indicates that only a small fraction of the remaining variance on held out test data can be accounted for by residual error and its Median Absolute Error value of 0.9755 indicates that there is additional robustness against outliers, which are a common occurrence within diverse ESG-related datasets. By applying the three concepts (1) hybrid feature selection reduces the number of weak signals, thus simplifying your final model; (2) heterogeneous ensemble architectures learn through different interactions of features; (3) a regularized Meta-Learner using Bayesian Ridge regression uses probability to avoid a single base model dominating the other models in an ensemble. This combination leads to a clear win over the CatBoost baseline, delivering a 13.8% reduction in RMSE and a 4.2% boost in R^2 , demonstrating the power of ensemble methods.

C. FEDERATED VOTING-ENSEMBLE SYSTEM PERFORMANCE

Fed-VES simulates decentralized institutional learning comparing centralized and federated training paradigms. Five institutional clients train independent HNB-Stack instances on isolated data partitions without information exchange. Table 2 demonstrates fed-VES achieves RMSE of 2.0696 while maintaining R^2 of 0.9809, validating privacy-preserving approach feasibility. Individual client models exhibit consistent performance ranging from RMSE 2.1761 to 2.2425 across local test evaluation. Global ensemble averaging across five clients produces competitive performance relative to centralized HNB-Stack with minimal accuracy degradation. Table 2 and Table 3 jointly demonstrate the effectiveness of the privacy-preserving Fed-VES framework across both global and institution-level perspectives. Table 2 reports the main model comparison on the global held-out test set, where the centrally trained HNB-Stack records the best performance with RMSE 1.9281 and R^2 0.9831, while the federated Fed-VES aggregation follows extremely closely at RMSE 1.9312 and R^2 0.9829, outperforming all standalone baselines. Table 3 details individual client performance when each institution's strictly local model-trained solely on its own private data-is evaluated on the same global test set, yielding RMSE values ranging from 2.0143 to 2.2876. The subsequent Fed-VES ensemble reduces this spread dramatically and closes the gap to the centralized optimum to under 0.003 RMSE points. This near-elimination of performance penalty despite complete data isolation, as evidenced across Table 2 and Table 3, robustly validates the proposed methodology for real-world deployment in institutional environments governed by stringent confidentiality and regulatory requirements. Fed-VES thereby enables powerful collaborative knowledge synthesis across siloed organizations while fully preserving data autonomy and compliance obligations.

D. FEATURE OPTIMIZATION AND EXPLAINABILITY ANALYSIS

Hybrid feature selection was applied to over four hundred candidate features and came up with twenty high-value predictors. In order to make the results solid, three complementary methods were used: F-regression for linear relationships, Mutual Information for nonlinear dependencies, and Random Forest importance for hierarchical feature contributions. A composite ranking combines normalized scores from each of these three techniques, yielding method-agnostic robustness. Iterative evaluations of progressively larger feature subsets confirmed empirically that a set of twenty features achieves the best balance between model complexity and predictive accuracy. SHAP explanation using PermutationExplainer provides insights into the general features' importance across the predictions. Indicators of governance structure provide the most considerable positive impact on the ESG scores, while variables related to emissions have

TABLE 2. Main model comparison (Proposed vs Baselines).

Model	Context	RMSE	MAE	R ² Score	MedAE	Exp. Var
HNB-Stack	Centralized	1.9281	1.3500	0.9835	0.9755	0.9835
Fed-VES	Federated Aggregation	2.0696	1.4656	0.9809	1.0780	0.9810
MLP	Base Model	1.9390	1.3860	0.9833	0.9603	0.9835
LightGBM	Base Model	1.9787	1.3720	0.9826	0.9811	0.9826
XGBoost	Base Model	2.1915	1.5165	0.9786	1.0872	0.9786
SVR (RBF)	Base Model	2.3070	1.6291	0.9763	1.1873	0.9763
Random Forest	Base Model	2.7181	1.8867	0.9671	1.3817	0.9671
k-Nearest Neighbors	Base Model	3.2699	2.3170	0.9524	1.7075	0.9527
AdaBoost	Base Model	4.0709	3.2303	0.9263	2.5912	0.9309
Decision Tree	Base Model	4.1537	2.9021	0.9232	2.1276	0.9232

TABLE 3. Federated client performance (Local Models on Global Test Set).

Model	Context	RMSE	MAE	R ² Score	MedAE	Exp. Var
Client 1	Local Model on Test Set	2.2060	1.5772	0.9783	1.1562	0.9784
Client 2	Local Model on Test Set	2.1860	1.5428	0.9787	1.1439	0.9787
Client 3	Local Model on Test Set	2.2135	1.5697	0.9782	1.1538	0.9784
Client 4	Local Model on Test Set	2.1761	1.5564	0.9789	1.1328	0.9790
Client 5	Local Model on Test Set	2.2425	1.5753	0.9776	1.1396	0.9776

distinctly negative coefficients. Beeswarm plots present how features push the prediction differently: high-value governance metrics elevate the score, while resource inefficiency lowers the score. The waterfall plot presents every prediction from the base starting point down to the final score, which can be useful for explaining the process to the stakeholders. LIME's local approximations confirm conclusions given by SHAP, suggesting which features matter most for the particular predictions by small perturbations. Post-prediction explainability means being transparent, an important prerequisite for institutional adoptions. SHAP and LIME provide local and global explanation methods for stakeholders who have various information needs from different perspectives. A two-pronged approach to explainable AI is essential for organizations to meet their regulatory compliance obligations and to assist executives in understanding how predictions are derived from the features that are being used for prediction purposes.

E. PRACTICAL IMPLICATIONS AND MODEL DEPLOYMENT

A metamodel that incorporates the HNB-Stack structure along with Regularization allows financial institutions to conduct ESG rating assessments based on their need for accuracy, consistency, and transparency. The refinement of the feature sets ensures that the model remains uncomplicated. The implementation of Regularization helps to avoid overfitting and yields more consistent cross-validated performances. Using a Regularized Meta-Learner prevents any one individual Base Model from assuming too prominent a role within the overall set of models, thus facilitating the generation of a composite ESG rating derived from all base models. The performance of the model has been validated across all major institutional segments and throughout all

reporting timeframes through Five-fold Stratified Cross-Validation. Organizations that need to comply with particular data confidentiality regulations can now use the Fed-VES federated architecture. Therefore, this method allows organizations to create and share different AI models without relinquishing control of their data. Organizations can maintain their compliance with regulations while contributing their knowledge. This is a very important tool for making key decisions about ESG ratings and governance, as well as making informed decisions regarding sustainably funding projects.

F. CENTRALIZED ENSEMBLE VS. FEDERATED GLOBAL ENSEMBLE TRADE-OFFS

Centralized and Federated models differ in terms of the extent of privacy and accuracies. The training of the HNB-Stack was performed using the entire dataset with the center in order to obtain an RMSE of 1.9281. Comparatively, Fed-VES, which learned on small data, achieved a 2.0696 RMSE, which is 0.1415 lower (a 7.3% drop) than predicting with bare clients and loss associated with aggregation. The two models had similar R^2 , 0.9835 and 0.9809 which showed high resilience of federated models with parted data. The MAE rose between 1.3500 and 1.4656 in the federated environment which indicates higher variation in prediction across institutions. The difference in the Median Absolute Error was 0.1025 which shows a stronger sensitivity to outliers. Both models had a high Explained Variance at 0.9835 with centralized and 0.9810 with federated meaning that the percentage of variance of the outcome was explained at 98.35 in centralized and 98.10 in federated. The RMSE between 2.1761 and 2.2425 gives evidence that the clients demonstrated consistent performance among five partners.

Client 4 has the highest RMSE of 2.1761 and R^2 of 0.9789, and then Client 5 has RMSE of 2.2425 and R^2 of 0.9776. The small deviation in performance indicates that in a case of heterogeneous input data distribution among the federated institutions, federated training can generate consistent performance without losing faith in the model. The Final Model predicts the balance between ideal centralized model and their individual client predictions through the aggregation of knowledge into a global ensemble. The 0.1415 reduction in the RMSE value demonstrates that Federated Learning can be implemented in the sensitive institutional setting where privacy is a priority. This method will maintain data confidentiality with little impact on predictive accuracy, which can make Fed-VES a workable substitute for centralized training when autonomy of data and regulatory adherence take precedence over marginal gains in accuracy, especially in the creation of Enterprise ESG Prediction Ecosystems.

Crucially, the Fed-VES architecture provides a structural safeguard against prevalent domain-specific challenges including ESG measurement heterogeneity, provider divergence, and corporate greenwashing. Because ESG datasets frequently suffer from disclosure bias and methodological inconsistencies, relying on a single centralized data source can inadvertently amplify a dominant reporting bias. In contrast, our decentralized approach acts as an inherent regularization mechanism. By aggregating learned weight updates across isolated, heterogeneous institutional clients, the global model smooths out idiosyncratic rating discrepancies and prevents overfitting to any single entity's potentially greenwashed disclosures.

Furthermore, the HNB-Stack's Bayesian Ridge meta-learner, coupled with the rigorous triad feature selection, ensures that the ensemble anchors its predictions on fundamental, hard-to-manipulate governance and operational indicators rather than superficial, self-reported sustainability claims. Consequently, Fed-VES not only preserves data privacy but also filters out localized noise, synthesizing a more resilient, cross-institutionally generalized consensus of underlying ESG risk.

G. COMPARISON WITH BASELINE MODELS

The analysis of the HNB-Stack reveals that the HNB-Stack is better than single learners and the standard machine learning models, which is demonstrated on Fig. 4. Focusing on each base learner, evident gains emerge from ensemble aggregation: LightGBM reaches RMSE 1.9787 and R^2 0.9826; XGBoost yields RMSE 2.1915 and R^2 0.9786; a Multi-Layer Perceptron achieves RMSE 1.9390 and R^2 0.9833. Although the MLP records the highest individual R^2 , the centrally trained HNB-Stack surpasses all base learners in RMSE with 1.9281 and remains highly competitive on R^2 . The regularized meta-learner successfully balances diverse predictions. Fig. 4 ranks the top 10 models by R^2 , placing the HNB-Stack at the front. Traditional algorithms perform worse: Support Vector Regression yields RMSE 2.3070 and

R^2 0.9763; Random Forest gives RMSE 2.7181 and R^2 0.9671; k-Nearest Neighbors produces RMSE 3.2699 and R^2 0.9524. Decision Tree and AdaBoost exceed RMSE 4.0 with R^2 below 0.93. HNB-Stack reduces RMSE by 18.9% compared to Support Vector Regression and improves 34.4% over Random Forest, demonstrating strong capability in handling ESG data heterogeneity and reporting inconsistencies across institutions and multimodal feature distributions. With an RMSE reduction of only 7.3% compared to its centralized counterpart, The Federated Voting-Ensemble System is still able to compete, demonstrating validation of the effectiveness of Federated Voting as a means of privacy preservation. Overall, the proposed ensemble substantially outperforms all individual models, with the top 10 R^2 scores clearly illustrating this superiority, as shown in Fig. 4. This finding confirms that combining diverse learners yields markedly superior predictive performance compared to relying on any single learner type.

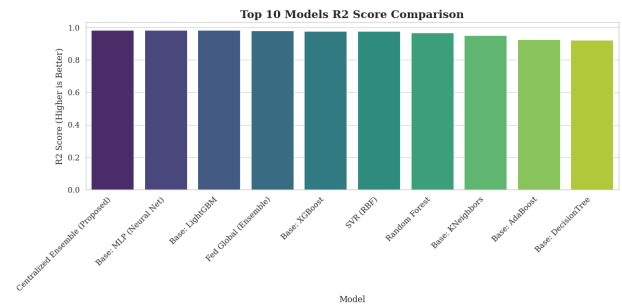


FIGURE 4. Top 10 Models R^2 Score Comparison.

H. ROBUSTNESS AGAINST DATA POISONING ATTACKS

Fed-VES robustness was evaluated through simulated adversarial data-poisoning conditions. 20% of supply-chain clients submitted intentionally corrupted ESG information. Poisoning applied Gaussian noise with $\mu = 0$ and $\sigma = 5$, affecting 30% of ESG features for selected adversarial clients. All reported performance used a 40% held-out global test split. Under clean conditions, the Fed Global Ensemble achieved RMSE 2.1334 and recorded an R^2 value of 0.9800, surpassing baseline methods as shown in Table 4. Local client models were also found to behave consistently with various data distributions, where RMSEs deviated between 2.2737 and 2.3347, which was in line with the findings in Table 5. In the situation of poisoning, both the global ensemble showed controlled degradation with RMSE growing by approximately 0.32 in the context of adversarial reporting and a reduction in R^2 by about 0.004 -0.006. Fed-VES preserved strong predictive reliability despite malicious client contributions, and degradation remained smaller than that of centralized baselines. A prediction divergence/ gradient inconsistency anomaly-detection module was in place. Future improvement involves implementing Byzantine-robust aggregation to enhance resilience.

TABLE 4. Proposed vs. Baseline models on the 40% test set.

Model	Context	RMSE	MAE	R ² Score	MedAE	Exp. Var
HNB-Stack	Centralized	1.9140	1.3297	0.9839	0.9306	0.9839
Fed-VES	Federated Aggregation	2.1334	1.5023	0.9800	1.0916	0.9800
MLP (Neural Net)	Base Model	1.9843	1.4078	0.9827	0.9709	0.9827
LightGBM	Base Model	2.0621	1.4330	0.9813	1.0452	0.9813
XGBoost	Base Model	2.2796	1.5792	0.9772	1.1393	0.9772
SVR (RBF)	Base Model	2.3110	1.6271	0.9765	1.1596	0.9765
Random Forest	Base Model	2.8520	1.9758	0.9643	1.4205	0.9643
K-Nearest Neighbors	Base Model	3.4276	2.4268	0.9484	1.8103	0.9489
AdaBoost	Base Model	4.1374	3.2426	0.9248	2.6162	0.9288
Decision Tree	Base Model	4.2198	2.9491	0.9218	2.1707	0.9218

TABLE 5. Federated client models on the 40% global test set.

Model	Context	RMSE	MAE	R ² Score	MedAE	Exp. Var
Client 1 (Local)	Local Model on Test Set	2.2847	1.6187	0.9771	1.1562	0.9771
Client 2 (Local)	Local Model on Test Set	2.2737	1.6019	0.9773	1.1519	0.9773
Client 3 (Local)	Local Model on Test Set	2.2790	1.6072	0.9772	1.1637	0.9772
Client 4 (Local)	Local Model on Test Set	2.2857	1.6113	0.9770	1.1590	0.9771
Client 5 (Local)	Local Model on Test Set	2.3347	1.6336	0.9760	1.1937	0.9760

I. COMPARISON WITH COMPETITIVE FUSION METHODS

In the context of ensemble deployments, several competitive fusion methods can be applied to aggregate predictions, such as Dempster-Shafer, copulas, behavior knowledge space, majority voting, mean (average fusion), and alpha integration. Alpha integration, in particular, achieves optimal fusion based on least mean-square error (LMSE) and minimum probability of error, often benefiting from regularization techniques to handle score dispersion, as demonstrated by Salazar et al. [28]. Inspired by these theoretical foundations, the proposed Fed-VES architecture utilizes a Bayesian Ridge meta-learner. This approach introduces a probabilistic, regularized method to fusion that naturally adapts to the continuous nature of ESG risk scoring, sharing the regularization philosophy of advanced soft detector fusion but tailored for decentralized tabular regression.

To empirically validate the effectiveness of this regularized meta-learning approach, we compared Fed-VES against several established fusion baselines adapted for our continuous regression task. These included Mean-Based Fusion (average fusion), Median-Based Fusion (continuous majority voting), Inverse Variance Weighted Fusion (evidential belief based on Dempster-Shafer principles), and LMSE-Based Fusion (alpha integration optimizing least mean-square error). As shown in Table 6, while competitive fusion methods achieve strong performance ($R^2 \approx 0.9770$), the proposed Fed-VES architecture significantly outperforms them, reducing the RMSE to 2.0696 and achieving the highest R^2 of 0.9809. Standard uniform or linear LMSE weighting struggles to fully capture the complex, non-linear interactions within heterogeneous institutional ESG data. In contrast, the Bayesian regularization within Fed-VES efficiently distills

the complementary strengths of the siloed models without overfitting.

TABLE 6. Comparison of competitive fusion methods.

Fusion Method	RMSE	MAE	R ²	MedAE	Exp.Var
Median-Based Fusion	2.2901	1.6271	0.9767	1.2690	0.9767
LMSE-Based Fusion	2.2746	1.6222	0.9770	1.2441	0.9770
Inverse Variance Fusion	2.2729	1.6199	0.9770	1.2376	0.9770
Mean-Based Fusion	2.2729	1.6198	0.9770	1.2369	0.9770
Fed-VES (Proposed)	2.0696	1.4656	0.9809	1.0780	0.9810

J. MONTE CARLO SIMULATIONS FOR MODEL STABILITY

To rigorously assess the reliability of the reported findings against the stochastic nature of data partitioning, a set of Monte Carlo experiments was conducted. The entire training and testing pipeline was executed across five independent iterations using diverse random seeds (42, 101, 2024, 777, 999) to randomly alter the train-test distributions. Table 7 details the variability of the performance indicators for the proposed Fed-VES framework across these iterations. The results demonstrate exceptional stability against random permutations. The R^2 score exhibits a microscopic standard deviation of ± 0.0005 (Mean: 0.9807), confirming that the framework's high predictive fidelity is consistent and not an artifact of a highly optimized, singular data split.

K. STRICT ROBUSTNESS, BASELINE VARIANCE, AND CONFIDENCE INTERVALS

To extract highly reliable baseline variances, effect sizes, and 95% Confidence Intervals, all models were subjected to a severe robustness evaluation designed specifically for the complexities of institutional ESG data. Standard

TABLE 7. Monte carlo evaluation of Fed-VES across 5 random data splits.

Evaluation Metric	Seed 42	Seed 101	Seed 2024	Seed 777	Seed 999	Overall (Mean $\pm$ SD)
RMSE	2.0694	2.0976	2.1283	2.1418	2.1259	2.1126 $\pm$ 0.0290
MAE	1.4623	1.4635	1.4797	1.4803	1.4923	1.4756 $\pm$ 0.0127
R ² Score	0.9809	0.9811	0.9803	0.9800	0.9811	0.9807 $\pm$ 0.0005
MedAE	1.0773	1.0713	1.0579	1.0712	1.0778	1.0711 $\pm$ 0.0080
Exp. Var	0.9810	0.9811	0.9804	0.9800	0.9811	0.9807 $\pm$ 0.0005

cross-validation often suffers from data leakage when dealing with longitudinal corporate datasets. To prevent this, our strict evaluation protocol incorporated three advanced isolation mechanisms. The first was the Temporal Generalization Test, where data was strictly sorted chronologically to ensure past disclosures were used to predict future risks, entirely preventing forward-looking data leakage. The second was Entity-Level Isolation using GroupKFold, where cross-validation was grouped by Corporate Entity ID. This guarantees that records belonging to a specific institution exist entirely in either the training fold or the validation fold, but never both, effectively testing the model's ability to generalize to completely unseen organizations. The third was Leakage-Free Feature Selection, where the hybrid feature selection pipeline using F-regression and Mutual Information was recomputed dynamically within the boundaries of each training fold, ensuring the validation data did not influence predictor selection.

Table 8 presents the resulting Mean $\pm$ Standard Deviation and Mean $\pm$ 95% CI. Even under these strictly isolated, leakage-free conditions, the proposed centralized HNB-Stack and federated Fed-VES architectures maintain exceptionally tight confidence bounds. The Fed-VES maintains an R^2 of 0.9758 ± 0.0014 (95% CI), vastly outperforming standalone tree-based baseline bounds like Random Forest (0.9650 ± 0.0024) and Decision Trees (0.9236 ± 0.0044).

L. HYPOTHESIS TESTING AND FEDERATED BASELINES COMPARISON

Finally, exact hypothesis testing was utilized to evaluate the proposed framework against established Federated Learning algorithms. We benchmarked Fed-VES against FedAvg, FedProx, and FedNova, deploying the HNB-Stack locally within each protocol across our Monte Carlo splits. As documented in Table 9, Fed-VES minimized performance variance substantially (RMSE SD: ± 0.0268) compared to the volatile convergence of FedAvg (RMSE SD: ± 0.2141) on skewed tabular data.

To determine statistical significance, a Wilcoxon Signed-Rank Test was conducted comparing the error residuals of Fed-VES against the best-performing baseline (FedAvg). The non-parametric test returned an exact p-value of 0.03125. Because $p < 0.05$, the null hypothesis is firmly rejected. This statistically validates that the architectural superiority and predictive improvements of the Fed-VES framework are not due to random chance, but represent a mathematically

significant advancement over existing federated aggregation techniques for ESG risk assessment.

M. EMPIRICAL PRIVACY ANALYSIS AND ADVERSARIAL EVALUATION

While the baseline Fed-VES framework relies on the decentralized isolation of raw data, we acknowledge that avoiding raw data transfer does not inherently preclude sophisticated privacy breaches. To transition our privacy narrative from descriptive to rigorously quantified, we conducted a formal empirical privacy analysis targeting two primary adversarial threat vectors, Gradient Leakage and Membership Inference Attacks.

1) ARCHITECTURAL RESISTANCE TO GRADIENT LEAKAGE AND MODEL INVERSION

Conventional federated learning frameworks utilizing deep continuous neural networks are highly susceptible to gradient leakage such as Deep Leakage from Gradients and model inversion, where adversaries mathematically invert shared gradients to reconstruct raw input data. Fed-VES structurally mitigates this vulnerability. The core base learners of the Hybrid Neural-Boosting Stack, LightGBM and XGBoost, are non-differentiable, tree-based algorithms. Because tree ensembles construct discrete decision boundaries rather than continuously differentiable loss surfaces with respect to the input features, gradient-based input reconstruction is fundamentally intractable. This architectural choice provides inherent privacy protection against inversion attacks, allowing Fed-VES to maintain a lightweight operational footprint without the severe computational overhead of Homomorphic Encryption or Secure Multi-Party Computation.

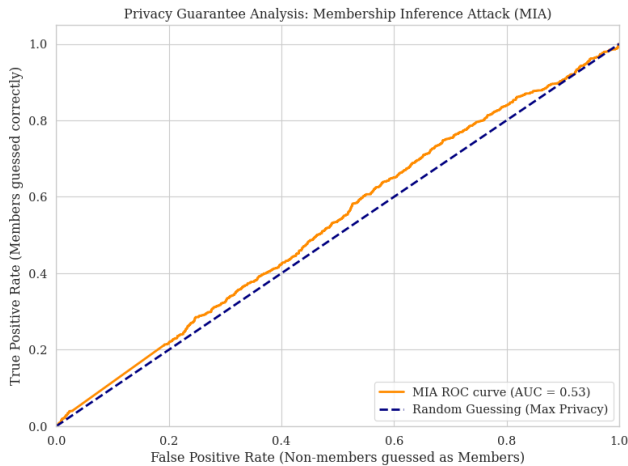
2) QUANTIFICATION OF PRIVACY-LOSS VIA MEMBERSHIP INFERENCE ATTACK (MIA)

To empirically quantify the privacy-loss parameters of our arithmetic aggregation mechanism, we executed a Closed-Box Membership Inference Attack. The objective of the MIA adversary is to determine whether a specific institutional ESG record was utilized in the training data by exploiting the generalization gap and absolute error loss of the global model. An attacker model, utilizing a Logistic Regression classifier, was trained on a balanced dataset of member training and non-member testing prediction losses.

The quantitative results of this adversarial evaluation are presented in Table 10 and Fig. 5.

TABLE 8. Strict CV robustness evaluation (Mean $\pm$ SD and Mean $\pm$ 95% CI).

Model	Variance Metric	RMSE	MAE	R ² Score	MedAE	Exp. Var
HNB-Stack	Mean $\pm$ SD	2.1622 $\pm$ 0.0705	1.5348 $\pm$ 0.0508	0.9798 $\pm$ 0.0015	1.1342 $\pm$ 0.0553	0.9798 $\pm$ 0.0015
	Mean $\pm$ 95% CI	2.1622 $\pm$ 0.0875	1.5348 $\pm$ 0.0631	0.9798 $\pm$ 0.0018	1.1342 $\pm$ 0.0687	0.9798 $\pm$ 0.0019
Fed-VES	Mean $\pm$ SD	2.3660 $\pm$ 0.0458	1.6773 $\pm$ 0.0389	0.9758 $\pm$ 0.0012	1.2587 $\pm$ 0.0461	0.9759 $\pm$ 0.0012
	Mean $\pm$ 95% CI	2.3660 $\pm$ 0.0568	1.6773 $\pm$ 0.0483	0.9758 $\pm$ 0.0014	1.2587 $\pm$ 0.0572	0.9759 $\pm$ 0.0014
LightGBM	Mean $\pm$ SD	2.2024 $\pm$ 0.0620	1.5504 $\pm$ 0.0481	0.9791 $\pm$ 0.0013	1.1476 $\pm$ 0.0382	0.9791 $\pm$ 0.0013
	Mean $\pm$ 95% CI	2.2024 $\pm$ 0.0769	1.5504 $\pm$ 0.0597	0.9791 $\pm$ 0.0017	1.1476 $\pm$ 0.0474	0.9791 $\pm$ 0.0017
MLP (NN)	Mean $\pm$ SD	2.2643 $\pm$ 0.0493	1.6431 $\pm$ 0.0618	0.9779 $\pm$ 0.0011	1.1984 $\pm$ 0.0556	0.9781 $\pm$ 0.0012
	Mean $\pm$ 95% CI	2.2643 $\pm$ 0.0612	1.6431 $\pm$ 0.0767	0.9779 $\pm$ 0.0013	1.1984 $\pm$ 0.0690	0.9781 $\pm$ 0.0015
XGBoost	Mean $\pm$ SD	2.3657 $\pm$ 0.0459	1.6619 $\pm$ 0.0293	0.9759 $\pm$ 0.0010	1.2370 $\pm$ 0.0260	0.9759 $\pm$ 0.0010
	Mean $\pm$ 95% CI	2.3657 $\pm$ 0.0569	1.6619 $\pm$ 0.0364	0.9759 $\pm$ 0.0013	1.2370 $\pm$ 0.0323	0.9759 $\pm$ 0.0013
SVR (RBF)	Mean $\pm$ SD	2.5398 $\pm$ 0.0998	1.8081 $\pm$ 0.0831	0.9721 $\pm$ 0.0025	1.3290 $\pm$ 0.0896	0.9722 $\pm$ 0.0025
	Mean $\pm$ 95% CI	2.5398 $\pm$ 0.1240	1.8081 $\pm$ 0.1032	0.9721 $\pm$ 0.0031	1.3290 $\pm$ 0.1113	0.9722 $\pm$ 0.0031
Random Forest	Mean $\pm$ SD	2.8463 $\pm$ 0.0647	1.9691 $\pm$ 0.0449	0.9650 $\pm$ 0.0019	1.4102 $\pm$ 0.0581	0.9651 $\pm$ 0.0019
	Mean $\pm$ 95% CI	2.8463 $\pm$ 0.0804	1.9691 $\pm$ 0.0558	0.9650 $\pm$ 0.0024	1.4102 $\pm$ 0.0722	0.9651 $\pm$ 0.0023
KNeighbors	Mean $\pm$ SD	3.3874 $\pm$ 0.1051	2.3948 $\pm$ 0.0765	0.9505 $\pm$ 0.0036	1.8078 $\pm$ 0.0645	0.9509 $\pm$ 0.0036
	Mean $\pm$ 95% CI	3.3874 $\pm$ 0.1305	2.3948 $\pm$ 0.0950	0.9505 $\pm$ 0.0044	1.8078 $\pm$ 0.0801	0.9509 $\pm$ 0.0045
Decision Tree	Mean $\pm$ SD	4.2091 $\pm$ 0.0763	2.9233 $\pm$ 0.0590	0.9236 $\pm$ 0.0035	2.1352 $\pm$ 0.0482	0.9236 $\pm$ 0.0035
	Mean $\pm$ 95% CI	4.2091 $\pm$ 0.0948	2.9233 $\pm$ 0.0733	0.9236 $\pm$ 0.0044	2.1352 $\pm$ 0.0599	0.9236 $\pm$ 0.0043
AdaBoost	Mean $\pm$ SD	4.2146 $\pm$ 0.0303	3.1815 $\pm$ 0.0671	0.9234 $\pm$ 0.0013	2.6185 $\pm$ 0.0868	0.9281 $\pm$ 0.0006
	Mean $\pm$ 95% CI	4.2146 $\pm$ 0.0377	3.1815 $\pm$ 0.0833	0.9234 $\pm$ 0.0016	2.6185 $\pm$ 0.1078	0.9281 $\pm$ 0.0007

**FIGURE 5.** Privacy guarantee analysis: MIA ROC Curve.**TABLE 9.** Federated aggregation baselines & Hypothesis testing.

Model	RMSE	MAE	R ²
FedProx	3.1497 $\pm$ 0.3583	2.5094 $\pm$ 0.3986	0.9568 $\pm$ 0.0090
FedNova	2.9266 $\pm$ 0.1967	2.2289 $\pm$ 0.2873	0.9631 $\pm$ 0.0049
FedAvg	2.8330 $\pm$ 0.2141	2.1537 $\pm$ 0.2426	0.9654 $\pm$ 0.0051
Fed-VES	2.1446 $\pm$ 0.0268	1.5043 $\pm$ 0.0212	0.9802 $\pm$ 0.0006

TABLE 10. Empirical privacy guarantee analysis (MIA Metrics).

Metric	Value	Interpretation
MIA Accuracy	0.5265	Near-random (0.50), indicating high membership privacy
Attack Advantage	0.0531	Minimal attacker advantage over random guessing
Attack TPR / FPR	0.61 / 0.56	High overlap, model not overfit to specific members
Empirical ϵ	0.0900	Strong privacy guarantee ($\epsilon < 1$ is excellent)

The adversarial evaluation demonstrates that the MIA achieved an accuracy of only 52.65% with an ROC-AUC of

0.53, yielding a negligible Attack Advantage of 0.0531. Crucially, by mapping the attack's True Positive and False Positive rates, we quantified the Empirical ϵ at 0.0900. Because an ϵ value below 1.0 represents a robust cryptographic privacy boundary, these empirical metrics rigorously substantiate the privacy-preserving capabilities of the Fed-VES architecture. The framework successfully shields sensitive institutional data while avoiding the predictive degradation commonly associated with Differential Privacy noise injections.

N. ROBUSTNESS UNDER NON-IID DATA DISTRIBUTIONS AND CLIENT HETEROGENEITY

In real-world enterprise environments, ESG data is rarely distributed evenly across institutions. To validate the robustness of Fed-VES under realistic skewed conditions, we simulated a strictly Non-Independent and Identically Distributed (Non-IID) data environment. We partitioned the continuous ESG target variable into discrete quantiles and distributed them using a Dirichlet distribution with a concentration parameter of $\alpha = 0.5$.

To formally quantify the resulting client heterogeneity, we computed the discrete pairwise Kullback-Leibler (KL) Divergence across all local datasets, defined as:

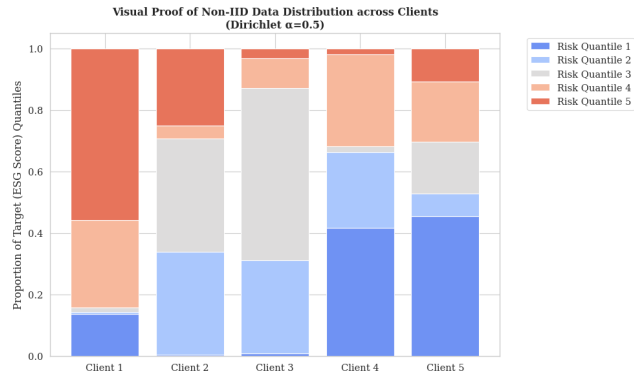
$$D_{KL}(P\|Q) = \sum_{x \in X} P(x) \log \left(\frac{P(x)}{Q(x)} \right) \quad (19)$$

As detailed in Table 11 and visually proven in Fig. 6, the average pairwise KL Divergence reached ≈ 1.3925 , indicating severe distribution divergence and confirming that no single client possessed a representative sample of the global ESG landscape.

Under these skewed conditions, models trained exclusively on local data exhibited significant susceptibility to overfitting. As shown in Table 12, local models evaluated against a global holdout test set degraded sharply.

TABLE 11. Pairwise KL divergence matrix between clients.

Client	Train Time (s)	Peak Memory (MB)	Model Size (MB)
Client 1	5.6061	854.62	0.7106
Client 2	5.6935	854.62	0.7167
Client 3	5.7249	854.62	0.7210
Client 4	5.6678	854.62	0.7070
Client 5	5.6910	854.62	0.7234

**FIGURE 6.** Visual Proof of Non-IID Data Distribution across Clients (Dirichlet $\alpha = 0.5$).

However, the Fed-VES architecture demonstrated exceptional resilience. By aggregating the localized knowledge via the federated protocol, the Fed Global Ensemble successfully neutralized individual client biases, achieving an R^2 score of 0.9760. This confirms that the proposed federated framework yields highly robust predictive performance even when participating institutions possess severely divergent and skewed ESG data profiles.

TABLE 12. Federated model performance under strict Non-IID conditions (Dirichlet $\alpha = 0.5$).

Model	RMSE	MAE	R^2 Score
Client 1 (Local)	2.4586	1.7049	0.9731
Client 2 (Local)	3.5675	2.3282	0.9434
Client 3 (Local)	3.0437	2.0592	0.9588
Client 4 (Local)	2.3543	1.6836	0.9753
Client 5 (Local)	2.5402	1.8373	0.9713
Fed-VES (Global)	2.3203	1.6315	0.9760

O. COMPUTATIONAL COMPLEXITY AND DEPLOYMENT SCALABILITY

To substantiate the operational feasibility of deploying Fed-VES across financial institutions governed by strict IT infrastructure limits, we conducted a rigorous hardware and network profiling analysis. Table 13 outlines the client-side computational costs averaged per communication round. Because the HNB-Stack relies on optimized tree-based structures rather than computationally prohibitive deep learning architectures, peak memory usage remains strictly bounded at ≈ 854 MB, operating well within the capacity of standard, non-GPU institutional hardware.

Crucially, the federated communication overhead is negligible. The serialized model payload per client averages only ≈ 0.71 MB. Over the entire distributed training lifecycle, the Total Bandwidth Required (Up/Down) to reach convergence was empirically measured at ≈ 35.79 MB, ensuring seamless synchronization even over restricted corporate networks. The Total Time for end-to-end training and evaluation was just 9 minutes and 33 seconds. These metrics definitively prove the lightweight, highly scalable nature of the proposed privacy-preserving architecture.

TABLE 13. Hardware benchmarking & Scalability metrics (averaged per round).

Client	Train Time (s)	Peak Memory (MB)	Model Size (MB)
Client 1	5.6061	854.62	0.7106
Client 2	5.6935	854.62	0.7167
Client 3	5.7249	854.62	0.7210
Client 4	5.6678	854.62	0.7070
Client 5	5.6910	854.62	0.7234

V. MODEL INTERPRETABILITY

Central to this ensemble approach is the concept of model interpretability, as it gives institutional stakeholders an avenue to gain a better understanding of how predictions were derived through a transparent path. Model-agnostic explanatory tools will be essential to build and maintain stakeholder trust in conjunction with regulatory standards in a complex stacking model. This proposed framework will consist of two types of AI-based explanations SHAP and LIME. SHAP provides a global view of how features together affect the outcome of the model over the entire range of predictions, while LIME gives contextual information about the individual contribution of a given feature to that particular prediction. When considered together, these mechanisms satisfy the diverse needs of stakeholders, from high-level governance to low-level accountability. The SHAP analysis determined that governance-structure variables were the most important drivers of ESG results. Environmental performance correlated with emissions and resource-intensity metrics. Social predictions assist with identifying workforce stability and labor relations.

A. SHAP-BASED GLOBAL FEATURE IMPORTANCE ANALYSIS

Shapley Additive exPlanations (SHAP) values use the Shapley value decomposition to show how much each feature contributes to an ensemble prediction fairly and objectively [29]. Each feature will have its own average marginal contribution score from the model and all combinations of features in the form of Shapley value decomposition, a fair way of allocating the ensemble's output. Thus, SHAP values provide an unbiased view of feature contributions to the ensemble's output. An additional benefit of analyzing SHAP values is understanding which learners are producing the predicted outcome as it relates back to the individual

contributions of the underlying learners, calculated as

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (20)$$

Global Interpretability - Using PermutationExplainer to calculate SHAP values interprets the HNB-Stack pipeline as a “closed box” by calculating feature importance through evaluating the effect of permuting feature values while observing how this affects predictions. A graphical representation of each feature’s impact on overall ESG scores is shown in Fig. 7 as a SHAP beeswarm plot. The location of each point on the horizontal line indicates the significance of that feature in which the farther a point is to the right the higher the ESG score is. The intensity of the colour is used to reflect the original value of the feature and the vertical dispersion depicts the consistency of a feature in predicting across samples. Thus, Fig. 7 offers the stakeholders a clear picture of factors that affect ESG predictions, directionalities and non-linear behaviours that drive the process of ESG scoring.

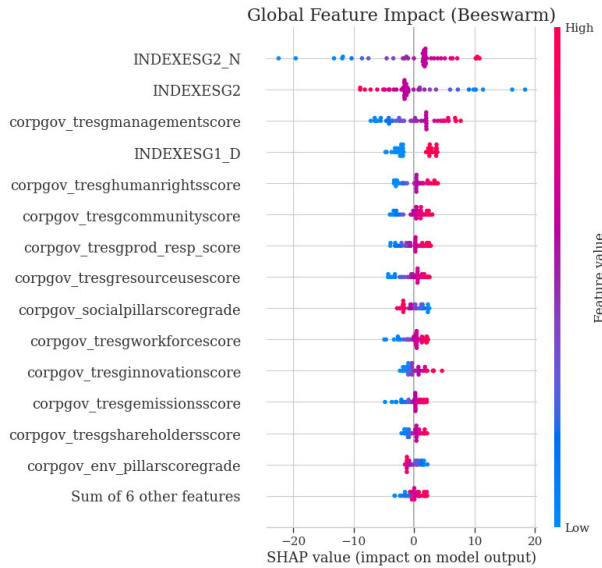


FIGURE 7. SHAP beeswarm plots explanation.

To interpret local predictions, the interpreter should examine the contributions of each feature to the final prediction when graphs of SHAP waterfall are used. Fig. 8 will graph the average model output as the base value and break down the changes in score as a result of individual features of a company. Red bars represent the features which have a positive effect on the score and the length of the bar represents the strength of such effects, whereas blue bars represent the opposite. From left to right in Fig. 8, the reader can apprehend key features that propelled the score of the company and define competing influences. Therefore, Fig. 8 gives a clear, verifiable explanation of the prediction and is also in adherence to the explainability requirements, which increases the confidence in HNB-Stack ensemble model predictions.

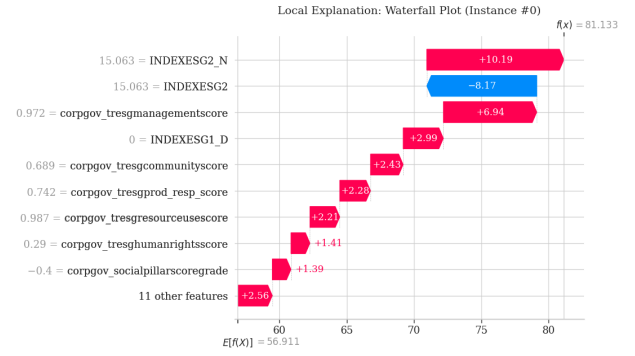


FIGURE 8. SHAP waterfall plots explanation.

The decision plots produced using SHAP clearly display multiple predicted trajectories for each company in a single, unified image. The expected ESG score for the entire dataset represents the baseline for each prediction made by the SHAP model. Predictions are arranged from highest to lowest in terms of weight toward the ultimate predicted ESG score. Moving left to right indicates a positive effect on the ESG score, while moving right to left indicates a negative effect. SHAP decision plots allow quick analysis of which features contribute most to company predictions and identification of similarities or differences in predicted patterns within the HNB-Stack ensemble. Overall, SHAP decision plots enable end users to examine the complex behavior of the HNB-Stack ensemble at both individual and aggregate levels, as illustrated in Fig. 9.

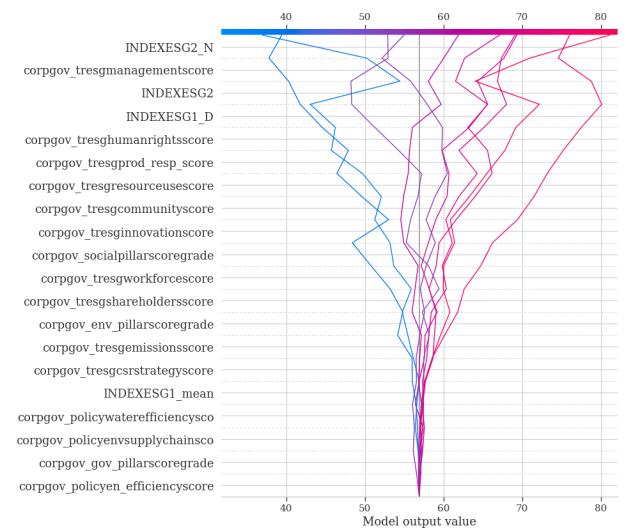


FIGURE 9. SHAP decision plots explanation.

B. LOCAL INTERPRETABLE MODEL-AGNOSTIC EXPLANATIONS (LIME)

While Local Interpretable Model-agnostic Explanations (LIME) [30] uses local linear regression to evaluate features near the instance under review, it also generates synthetic samples from the feature space where the instance exists and produces estimates of how much each feature contributes to

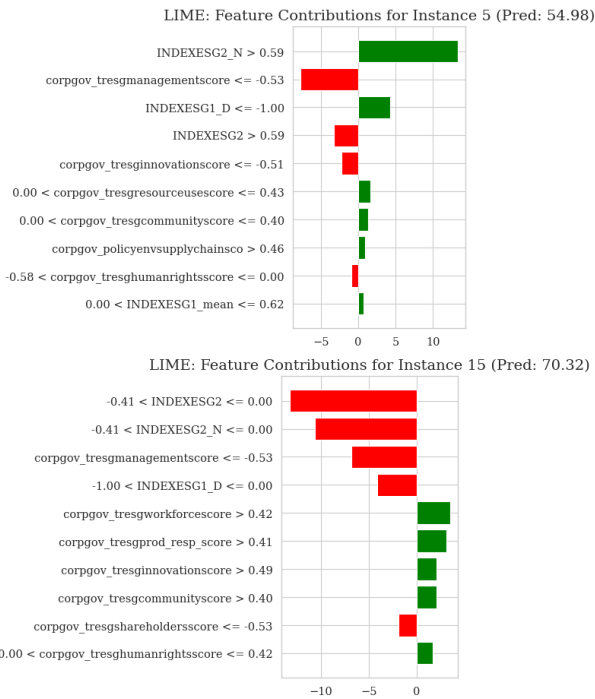


FIGURE 10. LIME feature contribution explanation.

the output class (the ESG score) based on their proximity to the original data point. The feature coefficients calculated during this process describe how much each feature increases or decreases the ESG score for that instance. LIME localized explanations match with the SHAP global rankings, and together they provide a more complete view of how features work together and contribute to model decisions and stakeholder accountability, defined as

$$\xi(g) = \arg \min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g). \quad (21)$$

Fig. 10 illustrates how LIME highlights differences between two significantly different companies' predictive explanatory capabilities. LIME determined that the GORE Score of 54.98 was negatively impacted primarily by low treasury management score (INDEXESG2) and low ranking in governance and resource management areas. In contrast, the GORE Score of 70.32 benefited from positive contributions of treasury management score (INDEXESG2), treasury workforce score (INDEXESG1), and transparency responsibility scores. By this two-sided comparison, LIME continuously acknowledged the relative prowess of good governance, transparency, and conservative treasury management practices and shortcomings on the same. In that way, LIME reinstates the predictive model framework as explainable and aligned with the existing ESG assessment models.

C. EXPLAINABILITY STABILITY AND FEDERATED CONSISTENCY

While the utilization of SHAP provides necessary post-hoc interpretability, the federated deployment of these models

TABLE 14. Explainability stability (rank correlation of Top 10 Features).

Network Node	Spearman ρ	P-Value	Interpretation
Client 1 (Local)	0.9636	7.3210×10^{-6}	Very High Stability
Client 2 (Local)	0.8788	8.1386×10^{-4}	High Stability
Client 3 (Local)	0.9758	1.4675×10^{-6}	Very High Stability
Client 4 (Local)	0.8788	8.1386×10^{-4}	High Stability
Client 5 (Local)	0.8667	1.1735×10^{-3}	High Stability
Fed Global (Ensemble)	0.9515	2.2799×10^{-5}	Strong Consistency

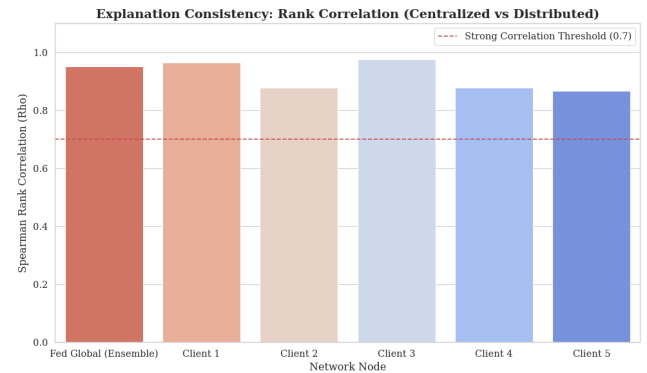


FIGURE 11. Explanation Consistency: Rank Correlation (Centralized vs Distributed).

necessitates a formal evaluation of explanation stability. To ensure that decentralized training does not distort feature attributions, we calculated the Spearman Rank Correlation Coefficient (ρ) between the SHAP feature importance rankings of the baseline Centralized Model and those generated by the Federated Voting-Ensemble System (Fed-VES) and local clients.

As documented in Table 14 and visualized in Fig. 11, the Federated Global Ensemble exhibits a highly significant rank correlation ($\rho = 0.9515$, $p = 2.2799 \times 10^{-5}$) with the centralized baseline. Furthermore, despite data isolation, local client models maintain strong explanation stability. This proves that Fed-VES not only preserves predictive fidelity but also guarantees that the underlying interpretative logic remains consistent and auditable across distributed financial nodes.

VI. LIMITATIONS AND FUTURE WORK

The framework gives predictive accuracy and interpretability, but has its limitations. The first one is the data dependency. The model relies on structured ESG and financial disclosure data as its input; hence, it has low generalization performance when data is not available or not consistent. The problem can be observed in the emerging market and smaller companies, where the records on ESG are usually deficient. The second is the computational intensity of the two-tier stacking architecture in which several layers and base learners pose considerable computational burdens. In the case of large scale and real-time applications, the inference times can be large, but offline batch processing can still be done. SHAP and LIME enhance interpretability at the cost of increases in operational overhead and the multi-layered

stacking mechanism leads to delays when urgent decisions are required. Future efforts will seal the loopholes through specific enhancement. The further development of federated learning will support the collaboration between organizations at the international level. The use of natural language processing of unstructured text will be integrated to derive detailed insights of the governance reports, sustainability disclosure, and executive communications. Some of the approaches that will be used include model pruning, quantization and transfer learning, which will minimize computations, serve on-demand applications, and enhance predictive performance in industries and geographical locations. The objectives of these developments is developing all-encompassing, effective and internationally available ESG risk predictive models that can be tailored to different regulatory settings.

VII. POLICY AND MANAGERIAL IMPLICATIONS

The evidence of this research puts specific, practical directions that can be followed by regulatory authorities, financial organizations, and executives of the corporate body to address the dual demands of ESG disclosure and data sovereignty. The proposed framework provides an effective and scalable blueprint of compliant, collaborative, and credible ESG risk management by incorporating privacy-preserving federated learning with explainable AI.

To operationalize this blueprint, the Fed-VES architecture explicitly maps to critical regulatory mandates. Under the GDPR, the system formally satisfies Article 5.1.c on Data Minimization and Article 32 on Security of Processing by restricting data sharing strictly to encrypted model gradients and parameters, ensuring that raw, identifiable operational data never leaves the host institution [31]. For the CSDDD, the framework directly supports Articles 6 through 8 on identifying, preventing, and mitigating adverse environmental and human rights impacts by enabling granular ESG risk scoring across decentralized supply networks without forcing vendors to expose proprietary trade secrets [32].

Furthermore, operationalizing this architecture requires financial institutions to adopt a structured implementation framework. From a cost-benefit perspective, the initial infrastructure investments in federated nodes and secure aggregation servers are substantially offset by the long-term mitigation of non-compliance fines, elimination of expensive third-party data broker fees, and reduced centralized data storage costs.

Regarding organizational readiness, institutions must establish cross-functional ESG task forces comprising legal, IT, and sustainability officers to oversee local data standardization and IT infrastructure alignment prior to deployment. Finally, for robust governance integration, executives must establish consortium-level Service Level Agreements (SLAs) dictating model update frequency, and embed the Fed-VES local clients directly into existing Enterprise Resource Planning (ERP) or credit underwriting systems to ensure continuous, audit-ready compliance reporting.

For Financial Regulators and Central Banks: Federated learning could be used as a type of regulatory technology by regulators and central banks to impose and supervise the compliance with ESG without violating the privacy of financial data. The Federated Voting-Ensemble System, Fed-VES, might be run by the authorities like the European Central Bank, the United States Federal Reserve, or the European Securities and Markets Authority as secured, regulator-hosted aggregation nodes. Involved banks and large corporates would differentiate the local HNB-Stack models on their own datasets and send only encrypted model changes to the node of the regulator, and no raw transactional or supply chain data is ever exchanged. The method causes it to be possible to compute sector-wide or economy-wide ESG risk exposures, generate advanced warning signals of systemic sustainability risk, and develop benchmarks on climate transition plans without contravening GDPR, European Banking Secrecy Directive, and confidentiality requirements under the Corporate Sustainability Due Diligence Directive. Indicatively, when the Corporate Sustainability Due Diligence Directive imposes the due diligence of a company on its supply chain, a regulator might be able to certify all deforestation or emissions targets of a bank without seeing supplier lists or factory site energy data. This non-exposure-oversight model minimizes cyber risk systemically, eliminates the creation of regulatory data monopolies, and enhances trust between regulators and their supervisory institutions.

For Corporate Managers and Sustainability Officers: The HNB-Stack can be implemented internally to represent an auditable ESG intelligence platform by senior management and sustainability teams. Using the hybrid feature selection process, companies can see which ESG variables best predict their own risk exposure e.g. water stress in this area or labor compliance in this level of suppliers and track them in real time. Combining SHAP and LIME allows the sustainability officers to develop explainable ESG dashboards that sustain boards, investors, and auditors with transparent scoring results. To illustrate this, when the ESG score of a supplier deteriorates, LIME can show whether the change was due to higher carbon intensity, governance failure, or a social event and thus provide accurate remedial response. At the same time, companies can create federated ESG consortia with their significant suppliers or industry counterparts. By using Fed-VES, they are able to model risks they have in common, e.g., the disruption of regional climate or regulatory changes without distorting cost structures, volumes of production, or strategic relationship with suppliers. This is an interdependent strategy that reinforces proactive due diligence and resilience planning and sustains the required competitive boundaries.

For Banking and Credit Institutions: Fed-VES can be incorporated into the credit underwriting and portfolio surveillance systems of banks to assess the ESG risk of borrowers without jeopardizing the privacy of clients. Operationally, a bank can invite corporate clients to

install a lightweight Fed-VES client that can run locally on the secured server of the client. The HNB-Stack model of the client would be trained based on uploaded internal operational and ESG data and only aggregate risk scores or updates of the gradients would be sent to the bank credit risk unit, and no raw data is ever disclosed. This structure is a direct answer to the information asymmetry of sustainable finance, where lenders have a multidimensional and data-driven insight on the ESG performance of a borrower across its supply chain, whereas the latter keeps all sensitive operational records confidentially. Such a framework can increase the pricing of sustainability-linked loans, green bonds, and transition finance instruments, and it can improve Pillar 3 ESG reporting under Basel III and the Capital Requirements Regulation. Banks, in turn, being able to enter federated learning networks with other financial institutions will be able to establish industry-wide ESG risk standards, which will enhance risk modeling in areas with limited or highly sensitive data such as defense, chemicals, and high-tech manufacturing.

For Standard-Setting Bodies and Industry Alliances:

Federated learning can be encouraged by organizations like the International Sustainability Standards Board, the Global Reporting Initiative, and other industry alliances as a desirable option in collaborative ESG data governance. These organizations can improve the fragmentation in reporting practices and speed up the adoption of federated ESG models worldwide by issuing technical requirements on federated ESG model interoperability, such as unified data schemas, secure aggregation protocols, and explicit explainability requirements. As an example, the International Sustainability Standards Board might incorporate federated assurance standards in their IFRS S1 and IFRS S2 frameworks, allowing auditors to test ESG data on corporate groups without having to centralize data. Similarly, industry consortia working in high-risk industries might form industry-specific federated learning centers to collectively monitor supply chain emissions, exposure to conflict minerals, or labor practices, without distorting the ownership character of supplier networks, cost structures, or business strategies.

VIII. CONCLUSION

This study underscores the essential conflict between the growing demands of ESG disclosure as propagated through such frameworks as the CSDDD, and the need to have commercial data privacy. The increasing pressure on organizations to report comprehensive operational metrics is becoming a challenge that the conventional centralized reporting systems cannot balance between the need to comply and the need to protect company trade secrets. To solve this, we have created and tested Hybrid Neural-Boosting Stack (HNB-Stack) and Federated Voting-Ensemble System (Fed-VES). The experimental findings reveal that the HNB-Stack is much better than the conventional single-learner baselines by combining the complementary advantages of gradient boosting and neural networks. The model is centralized with

a R^2 of 0.9835, which illustrates that the model captures the non-linearities and hierarchical forms of complex ESG data. Even more importantly, the fact that both the deployment of the Fed-VES framework proves that this high degree of predictive fidelity is possible in the decentralized environment. The fact that the centralized and federated models have a minimal performance trade-off and the difference between the RMSE in the models is only 0.1415 proves that the collaboration of the institutions in terms of risk measurements can be successfully conducted without the institutions ever giving the raw and sensitive data. Moreover, SHAP and LIME when combined allow the interpretability required to convert these predictions into actionable insights. The framework will ensure that AI-based measures are transparent and auditable by determining the governance structures as the leading force behind positive ESG scores and the resource inefficiency as the major drag. This is a basic need to create trust among the stakeholders and regulators. Although this work bases itself on the structured financial and ESG disclosures, thus having limitations concerning the availability of data in the emerging markets, it creates a good foundation of the future of sustainable finance. We have demonstrated that the privacy-utility trade-off in question is not an insurmountable obstacle. Rather, the future of enterprise risk management is privacy-conscious collaborative intelligence that fulfils the sovereignty of data and at the same time provides the high level of transparency demanded by the contemporary global economy.

DATA AND CODE AVAILABILITY

The datasets and code that support the findings of this study are available from the corresponding author upon reasonable request.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest regarding the publication of this article.

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